Section C will contain extended answer questions on contemporary contexts. This may contain stimulus materials on a scenario that students must read in order to answer the questions. It will focus on the chemistry behind the contexts and will not be a comprehension exercise. The longer timing of the examination reflects the style of the questions for Section C.

Students will be able to show their full ability in Sections B and C as these will contain areas where they will be stretched and challenged.

Students may use a calculator.

The quality of written communication will be assessed in the context of this unit through questions which are labelled with an asterisk (*). When answering these questions students should consider spelling, punctuation and grammar of their response, as well as the clarity of expression.

5.3 Redox and the chemistry of the transition metals

Students will be assessed on their ability to:

1 Application of redox equilibria

a demonstrate an understanding of the terms *oxidation number*, *redox*, *half-reactions* and use these to interpret reactions involving electron transfer

b relate changes in oxidation number to reaction stoichiometry

c recall the definition of standard electrode potential and standard hydrogen electrode and understand the need for a reference electrode

d set up some simple cells and calculate values of $E_{cell}^{\ominus}$ from standard electrode potential values and use them to predict the thermodynamic feasibility and extent of reactions

e demonstrate an understanding that $E_{cell}^{\ominus}$ is directly proportional to the total entropy change and to $\ln K$ for a reaction

f demonstrate an understanding of why the predictions in 5.3.1d may not be borne out in practice due to kinetic effects and non-standard conditions

g carry out and evaluate the results of an experiment involving the use of standard electrode potentials to predict the feasibility of a reaction, e.g. interchange of the oxidation states of vanadium or manganese

h demonstrate an understanding of the procedures of the redox titrations below (i and ii) and carry out a redox titration with one:

- i potassium manganate(VII), e.g. the estimation of iron in iron tablets
- ii sodium thiosulfate and iodine, e.g. estimation of percentage of copper in an alloy

i discuss the uncertainty of measurements and their implications for the validity of the final results

j discuss the use of hydrogen and alcohol fuel cells as energy sources, including the source of the hydrogen and alcohol, e.g. used in space exploration, in electric cars

k demonstrate an understanding of the principles of modern breathalysers based on an ethanol fuel cell and compare this to methods based on the use of IR and to the reduction of chromium compounds.

2 Transition metals and their chemistry

a describe transition metals as those elements which form one or more stable ions which have incompletely filled d orbitals

b derive the electronic configuration of the atoms of the d-block elements (Sc to Zn) and their simple ions from their atomic number

c discuss the evidence for the electronic configurations of the elements Sc to Zn based on successive ionisation energies

d recall that transition elements in general:

- i show variable oxidation number in their compounds, e.g. redox reactions of vanadium
- ii form coloured ions in solution
- iii form complex ions involving monodentate and bidentate ligands
- iv can act as catalysts both as the elements and as their compounds

e recall the shapes of complex ions limited to linear $[CuCl_2]^-$, planar $[Pt(NH_3)_2Cl_2]$, tetrahedral $[CrCl_4]^-$ and octahedral $[Cr(NH_3)_6]^{3+}$, $[Cu(H_2O)_6]^{2+}$ and other aqua complexes

f use the chemistries of chromium and copper to illustrate and explain some properties of transition metals as follows:

- i the formation of a range of compounds in which they are present in different oxidation states
- ii the presence of dative covalent bonding in complex ions, including the aqua-ions
- iii the colour or lack of colour of aqueous ions and other complex ions, resulting from the splitting of the energy levels of the d orbitals by ligands
- iv simple ligand exchange reactions
- v relate relative stability of complex ions to the entropy changes of ligand exchange reactions involving polydentate ligands (qualitatively only), e.g. EDTA
- vi relate disproportionation reactions to standard electrode potentials and hence to $E_{cell}^{\ominus}$

g carry out experiments to:

   i investigate ligand exchange in copper complexes

   ii study the redox chemistry of chromium in oxidation states Cr(VI), Cr(III) and Cr(II)

   iii prepare a sample of a complex, e.g. chromium(II) ethanoate

h recall that transition metals and their compounds are important as catalysts and that their activity may be associated with variable oxidation states of the elements or surface activity, e.g. catalytic converters in car exhausts

i explain why the development of new catalysts is a priority area for chemical research today and, in this context, explain how the scientific community reports and validates new discoveries and explanations, e.g. the development of new catalysts for making ethanoic acid from methanol and carbon monoxide with a high atom economy (green chemistry)

j carry out and interpret the reactions of transition metal ions with aqueous sodium hydroxide and aqueous ammonia, both in excess, limited to reactions with aqueous solutions of Cr(III), Mn(II), Fe(II), Fe(III), Ni(II), Cu(II), Zn(II)

k write ionic equations to show the difference between amphoteric behaviour and ligand exchange in the reactions in 5.3.2g

l discuss the uses of transition metals and/or their compounds, e.g. in polychromic sun glasses, chemotherapy drugs.